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# EXPERIMENT NO-8

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**Study and construction a 4 to 1 multiplexer using  
fundamental (basic) logic gates.**



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## Experiment No – 8, Study and construction a 4 to 1 multiplexer using fundamental (basic) logic gates.

### Theory:

A multiplexer, often abbreviated as "mux," is a fundamental digital electronic device used in computer systems and communication circuits. Its primary function is to select one of several input signals and route it to a single output line based on the control signals it receives. This process allows multiple data inputs to be transmitted through a single channel or line.

The basic concept of a multiplexer involves multiple data inputs, a selection control input (or inputs), and a single output. The selection control inputs determine which input signal is transmitted to the output.

Let the input line be  $m$  and the select lines be  $n$ , then the relation between the select lines and the input lines are given as

$$2^n = m \quad (1)$$

The block diagram of the 4 to 1 multiplexer is shown in the figure (8.1).

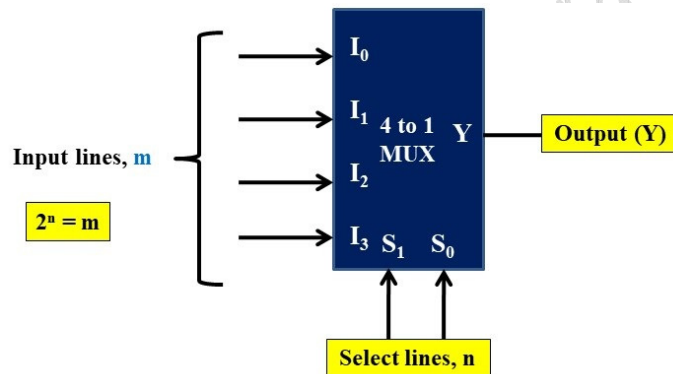


Figure (8.1), block diagram of a 4 to 1 multiplexer.

The truth table for a 4-to-1 multiplexer is depicted in Table (8.1), where the select lines assume a crucial role by accepting digital inputs and determining the chosen input directed to the output.

When the select line carries a logic zero, the output aligns with input  $I_0$ . Likewise, the functional table in Table (8.2) succinctly outlines the output's selection based on various combinations of select lines.

| Logical Select Inputs |       | Logical outputs |
|-----------------------|-------|-----------------|
| $S_1$                 | $S_0$ | Y               |
| 0                     | 0     | $I_0$           |
| 0                     | 1     | $I_1$           |
| 1                     | 0     | $I_2$           |
| 1                     | 1     | $I_3$           |

Table 8.1, The truth table / functional table of a 4 to 1 multiplexer.

The characteristic expression can be derived from the functional table (8.1).

Since, the outputs are given as the input lines to the

$$Y = I_0 \cdot \overline{S_1} \cdot \overline{S_0} + I_1 \cdot \overline{S_1} \cdot S_0 + I_2 \cdot S_1 \cdot \overline{S_0} + I_3 \cdot S_1 \cdot S_0 \dots\dots\dots (1)$$

The Boolean expression in the equation (1), can be implemented using fundamental or the basic logic gates as shown in the figure (8.2).

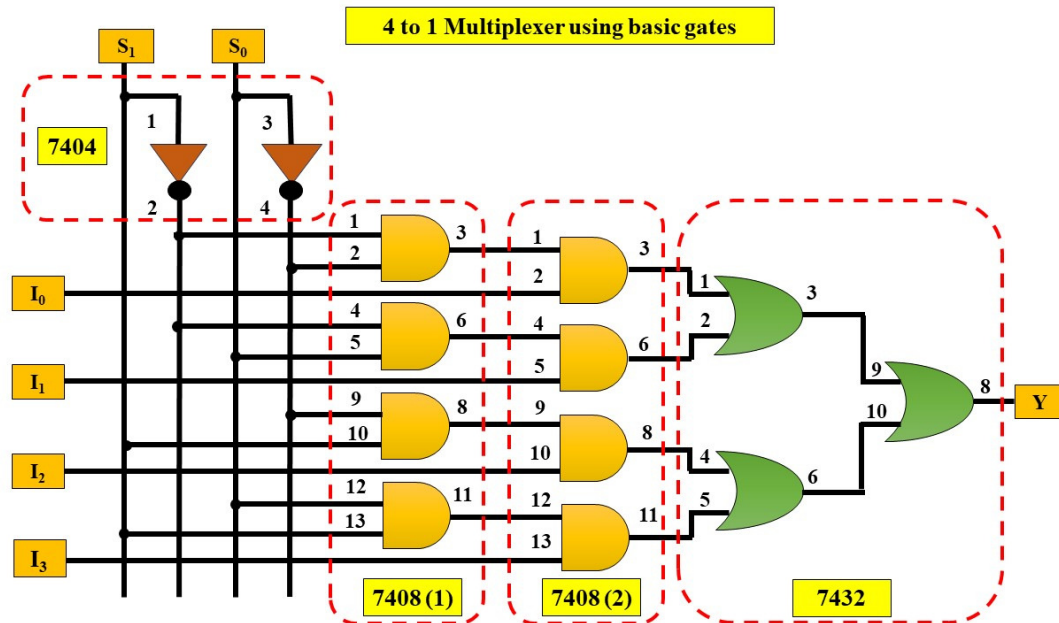


Figure (8.2), Logic circuit diagram of 4 to 1 multiplexer using basic gates.

### Semiconductors and Instruments required

#### Power Supply

1. DC +5V regulated power supply.

#### Integrated Circuits

1. Quad 2-input AND gate (7408)
2. Quad 2-input OR gate (7432)
3. Hex inverter, NOT Gate (7404)

#### Measuring Scientific equipment's

1. Digital Multimeter

#### Miscellaneous

1. Wish Board / Bread board
2. Jumper / connecting wires

## Result/Observation

Truth table for the 4 to 1 multiplexer using fundamental / basic and other logic gates.

| Select Input   |                | Logical Input  |                |                |                | Output      |             |
|----------------|----------------|----------------|----------------|----------------|----------------|-------------|-------------|
| S <sub>1</sub> | S <sub>0</sub> | I <sub>0</sub> | I <sub>1</sub> | I <sub>2</sub> | I <sub>3</sub> | Logical (Y) | Voltage (Y) |
| 0              | 0              | 0              | X              | X              | X              | 0           |             |
| 0              | 0              | 1              | X              | X              | X              | 1           |             |
| 0              | 1              | X              | 0              | X              | X              | 0           |             |
| 0              | 1              | X              | 1              | X              | X              | 1           |             |
| 1              | 0              | X              | X              | 0              | X              | 0           |             |
| 1              | 0              | X              | X              | 1              | X              | 1           |             |
| 1              | 1              | X              | X              | X              | 0              | 0           |             |
| 1              | 1              | X              | X              | X              | 1              | 1           |             |

Table (8.2), Experimental observation of 4 to 1 multiplexer.

## Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V** (V<sub>CC</sub>) or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.
6. In this circuit we are implementing using OR, AND & NOT logic gates.